Thermal radiation and color

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use peak wavelength as one detail that helps explain a thermal spectrum.

SUCCESS CRITERIA - I can...

1. Relate higher temperature to shorter peak wavelength.
2. Use kelvin in Wien's law.
3. Distinguish spectral peak from perceived color.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use "blackbody" and "peak" to explain your answer. Sentence starters are optional.

blackbody: an ideal absorber/emitter used in a thermal-radiation model

peak: the location of the largest value on a specified graph

kelvin: the absolute temperature scale used in thermal ratios

01 Warm-up: bring back an earlier idea

A light ray strikes a plane mirror at **20°** from the normal. **Find the reflection angle** and **label both angles** on a sketch.

Word help: reflection = a wave returning from a boundary

Sentence starter: My design includes ... . It does what is needed by ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $\lambda_{max}\, T = \mathbf{2.9\times10^{-3}\ m\cdot K}$.
Ideal blackbody; T in kelvin, λ_{max} in meters. A shorter peak wavelength means higher temperature, not one exact visible color.

Worked reasoning: A thermal spectrum's peak shifts toward shorter wavelength as absolute temperature rises. Wien's relation $\lambda_{peak}\, T \approx \mathbf{2.9\times10^{-3}\ m\cdot K}$ gives $\lambda_{peak} \approx \mathbf{9.7}$ micrometers at **300 K**.

Your turn: Sketch two broad thermal spectra, labeling axes and the hotter peak's shift.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .

Relative spectral emission



Wavelength (label units and peaks)

Marzano check - circle: ① ② ③

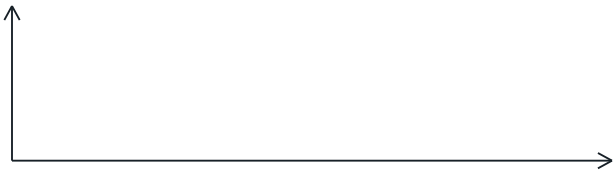
03 Read and explain the diagram or equation

Two ideal thermal emitters are at **3000** and **6000 K**. Which has the shorter peak wavelength? Does that mean they emit only one color?

Word help: ideal = using stated simplifying conditions

Sentence starter: I notice ... . This supports ... because ... .

Relative spectral emission



Wavelength (label units and peaks)

Marzano check - circle: ① ② ③

04 Apply the physics

Two calculated thermal-emission spectra have peaks at about **970 nm** for **3000 K** and **480 nm** for **6000 K**.
1. Draw and **label two broad spectra**.
2. **Compare peak wavelengths** and **name the peak outside the** approximate visible interval, **400**-**700 nm**.

Sentence starter: Both ... . They differ in ... because ... .

Relative spectral emission



Wavelength (label units and peaks)

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

An object has an infrared peak but looks red-hot. **Explain how both can be** true.

Sentence starter: I think … . I can point to …, which shows …, so … .

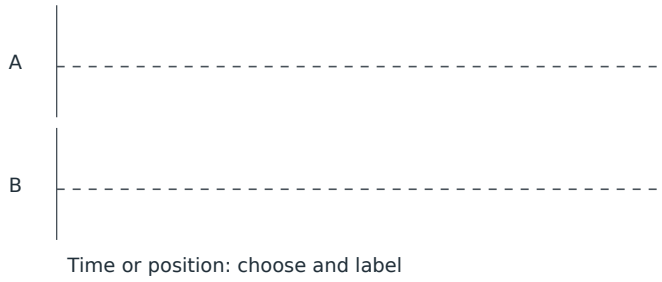
Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

An upward rope pulse is **4 cm** high before reaching an ideal free end. Is the reflected displacement upright or inverted? **Sketch it**.
Label a wave sketch to **support the calculation or comparison**.

Word help: ideal = using stated simplifying conditions

Sentence starter: In my work, I used … to show … . I labeled it … .



Time or position: choose and label

Marzano check - circle: 1 2 3

07 Exit check

Absolute temperature doubles. How does peak wavelength change?

Sentence starter: I notice … . This supports … because … .

Relative spectral emission



Wavelength (label units and peaks)

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

If temperature doubles, compare peak wavelengths. **Explain why a room-temperature object** can radiate even when it is not visibly glowing.

Sentence starter: Both … . They differ in … because … .

Relative spectral emission



Wavelength (label units and peaks)

Marzano check - circle: 1 2 3